

Problem

For the circuit in Fig. 7.88, if

$$v = 10e^{-4t} \text{ V} \quad \text{and} \quad i = 0.2e^{-4t} \text{ A}, \quad t > 0$$

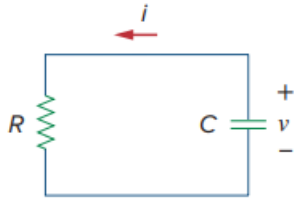
(a) Find R and C .

(b) Determine the time constant.

(c) Calculate the initial energy in the capacitor.

(d) Obtain the time it takes to dissipate 50 percent of the initial energy.

Fig. 7.88:



Step-by-step solution

Step 1 of 7

(a)

Refer to Figure 7.88 in the textbook.

Consider the voltage across capacitor in series RC circuit is,

$$v = 10e^{-4t} \text{ V} \quad \dots\dots (1)$$

Consider the general equation for the voltage across capacitor.

$$v = Ae^{-\left(\frac{t}{\tau}\right)} \quad \dots\dots (2)$$

Compare equation 2 with equation 1.

$$A = 10$$

$$\frac{t}{\tau} = 4t$$

$$\tau = \frac{1}{4}$$

Step 2 of 7

Consider the current through capacitor is,

$$i = 0.2e^{-4t}$$

Consider the equation for current through capacitor.

$$i = C \frac{dv}{dt} \dots\dots (3)$$

Replace $0.2e^{-4t}$ for i , $10e^{-4t}$ for v in equation 3.

$$-0.2e^{-4t} = C \frac{d}{dt}(10e^{-4t})$$

$$-0.2e^{-4t} = C(10)(-4e^{-4t})$$

$$-0.2 = -40C$$

$$C = 5 \text{ mF}$$

[Comments \(1\)](#)



Anonymous

Why is i negative in the equation

Step 3 of 7

Consider the equation for time constant τ .

$$\tau = RC \dots\dots (4)$$

Replace $\frac{1}{4}$ for τ , 5 mF for C in equation 4.

$$\tau = RC$$

$$\frac{1}{4} = R(5 \text{ mF})$$

$$R = \frac{1}{4(5 \text{ mF})}$$

$$R = 50 \Omega$$

Therefore, the value of R is and the value of C is .

Step 4 of 7

(b)

Determine the time constant τ .

Replace 50Ω for R , and 5 mF for C in equation 4.

$$\tau = RC$$

$$= (50)(5 \text{ mF})$$

$$= 250 \text{ ms}$$

Therefore, the value of the time constant τ is .

Step 5 of 7

(c)

Determine the initial energy in the capacitor $W_C(0)$.

Consider the equation for $W_C(0)$.

$$W_C(0) = \frac{1}{2}(C)(v_o^2) \dots\dots (5)$$

Replace $t = 0$ in equation 1.

$$\begin{aligned} v_o &= 10e^{-4t} \text{ V} \\ &= 10e^{-4(0)} \\ &= 10 \end{aligned}$$

Replace 10 for v_o , and 5 mF for C in equation 5.

$$\begin{aligned} W_C(0) &= \frac{1}{2}(5 \times 10^{-3})(10)^2 \\ &= \frac{1}{2}(5 \times 10^{-3})100 \\ &= 0.25 \text{ J} \end{aligned}$$

Step 6 of 7

(d)

Determine the time taken to dissipate 50 percent of its initial energy.

Consider the equation for energy dissipated by resistor.

$$W_R = \frac{1}{2}(C)(v_o^2) \left[1 - e^{\left(\frac{-2t_o}{\tau} \right)} \right] \dots\dots (6)$$

$$\begin{aligned} W_R &= \frac{50}{100}(250 \text{ mJ}) \\ &= 0.125 \end{aligned}$$

[Comments \(1\)](#)



Anonymous

This only calculated for 50% of the initial energy, not the time taken to get this value.

Step 7 of 7

Replace 10 for v_o , 0.125 for W_R , 250 ms for τ , and 5 mF for C in equation 6.

$$0.125 = \frac{1}{2}(5 \times 10^{-3})(100) \left[1 - e^{\left(\frac{-2t_o}{(250 \times 10^{-3})} \right)} \right]$$

$$0.5 = \left[1 - e^{(-8t_o)} \right]$$

$$e^{(-8t_o)} = 0.5$$

$$t_o = -\frac{1}{8} \ln(0.5)$$

$$\begin{aligned} t_o &= -\frac{1}{8} \ln(0.5) \\ &= 0.0866 \text{ s} \end{aligned}$$

Therefore, the time taken to dissipate 50 percent of the initial energy is 0.0866 s.